

ACADEMIC PROJECT ABSTRACT

CAR SHOWROOM

An Online Car Showroom Web Site

PROJECT NAME	CAR SHOWROOM – CAR SHOWROOM MANAGEMENT SYSTEM (CSMS)
CLIENT-SIDE	HTML5, CSS3, JavaScript
TECHNOLOGY	PHP
DATABASE	MySQL
SUBMITTED BY	HARSHIL VEKARIYA
ACADEMIC YEAR	2026-27

1. Project Abstract

Car Showroom is a dynamic, database-driven web application designed to showcase, manage, and sell a curated inventory of cars online for a modern car showroom. Built using PHP for server-side logic and MySQL for persistent data storage, the system bridges the gap between car dealerships and customers by offering a seamless digital shopping and test-drive booking experience.

The platform enables customers to browse cars by brand and category, view detailed specifications, add vehicles to a shopping cart, and place orders or book test drives securely. On the administrative side, the system provides a dedicated dashboard for managing the vehicle catalogue, tracking orders, and maintaining customer records — all through an intuitive, role-based interface.

This project demonstrates the practical application of core web development concepts including client-server architecture, relational database design, CRUD operations, session handling, and form validation, making it a comprehensive academic showcase of full-stack web development using the PHP–MySQL stack.

2. Description of the Website

Purpose

Car Showroom aims to digitize the traditional car-showroom experience by providing an organized, searchable, and visually appealing online catalogue of vehicles, along with a streamlined ordering, test-drive booking, and administrative workflow.

Key Modules

- **User Module** – Registration, login, profile management, and secure authentication.
- **Catalogue Module** – Browse cars by category/brand, view specifications, pricing, and stock availability.
- **Cart & Order Module** – Add to cart, update quantities, checkout, and track order status.
- **Admin Module** – Add, update, or remove cars; manage categories; view and process customer orders.
- **Reporting Module** – Generate sales and inventory reports for business insights.

3. Folder Structure

The project follows a modular folder structure that separates administrative, user-facing, and core functionalities, ensuring maintainability and scalability.

```
CarShowroom/  
|-- admin/  
| |-- dashboard.php  
| |-- manage_cars.php  
| |-- manage_category.php  
| |-- view_orders.php  
| `-- reports.php  
|-- user/  
| |-- register.php  
| |-- login.php  
| |-- profile.php  
| |-- cart.php  
| `-- order.php  
|-- includes/  
| |-- db_connect.php  
| |-- header.php  
| |-- footer.php  
| `-- functions.php  
|-- assets/  
| |-- css/style.css  
| |-- js/script.js  
| `-- images/ (car images & icons)  
|-- config/  
| `-- config.php  
|-- index.php  
|-- product_details.php  
|-- checkout.php  
|-- logout.php  
`-- database/  
    `-- car_showroom.sql
```

4. Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how information moves between the Customer, Admin, system processes, and the underlying data stores within Car Showroom.

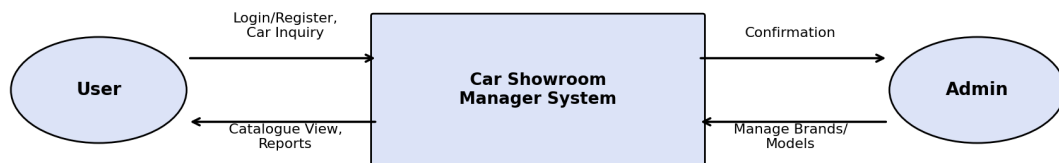


Figure 1: Level-1 Data Flow Diagram of Carzify

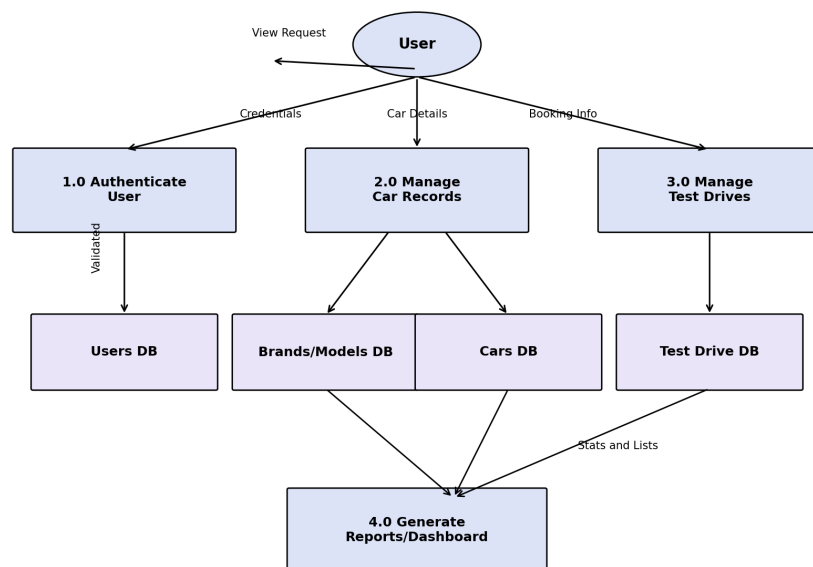


Figure 1b: Level-2 Data Flow Diagram of Carzify

5. Entity-Relationship (ER) Diagram

The ER Diagram below depicts the relational database design of Car Showroom, highlighting Primary Keys (PK), Foreign Keys (FK), and the relationships (1:M) between entities.

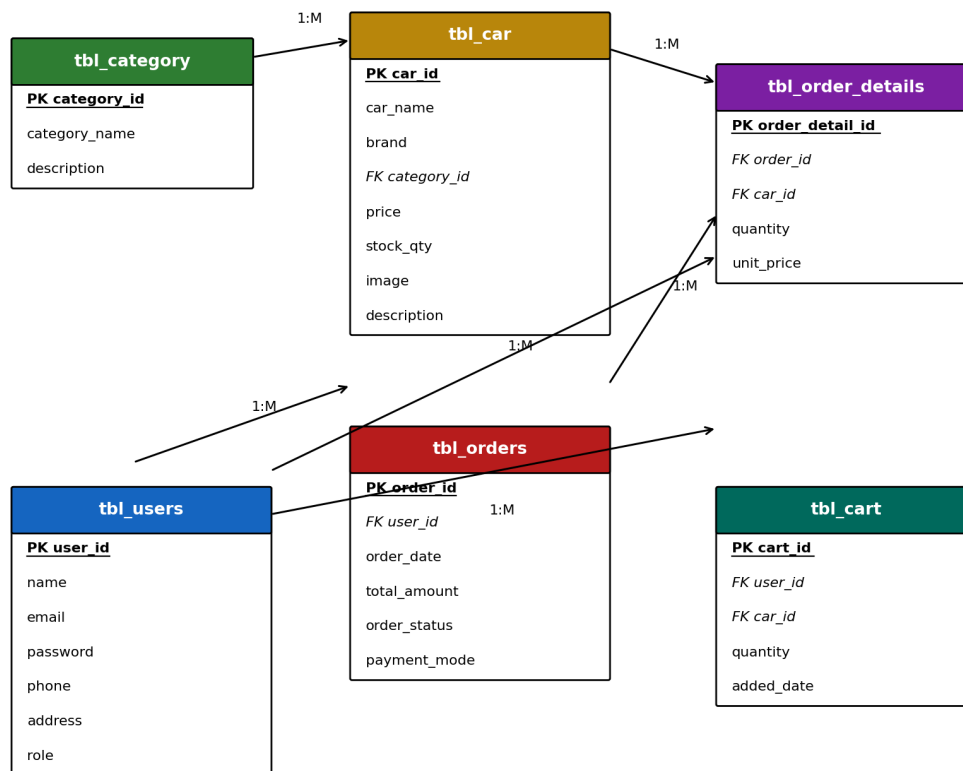


Figure 2: Entity-Relationship Diagram of Carzify Database

6. Data Dictionary

The data dictionary defines the structure of each table used in the Car Showroom database, specifying field names, data types, sizes, and key constraints (Primary Key / Foreign Key).

Table: tbl_users

Field Name	Data Type	Size	Key	Description
user_id	INT	11	PK	Unique identifier for each user (Auto Increment)
name	VARCHAR	100	-	Full name of the user
email	VARCHAR	100	-	Unique email address used for login
password	VARCHAR	255	-	Encrypted user password
phone	VARCHAR	15	-	Contact number of the user
address	TEXT	-	-	Delivery / billing address
role	VARCHAR	20	-	Defines user type: Customer or Admin

Table: tbl_category

Field Name	Data Type	Size	Key	Description
category_id	INT	11	PK	Unique identifier for each category (Auto Increment)
category_name	VARCHAR	100	-	Name of the car category (e.g. Sedan, SUV, Hatchback)
description	TEXT	-	-	Brief description of the category

Table: tbl_car

Field Name	Data Type	Size	Key	Description
car_id	INT	11	PK	Unique identifier for each car (Auto Increment)
car_name	VARCHAR	150	-	Name/model of the car
brand	VARCHAR	100	-	Brand or manufacturer name
category_id	INT	11	FK	References category_id in tbl_category
price	DECIMAL	10,2	-	Selling price of the car
stock_qty	INT	11	-	Available stock quantity
image	VARCHAR	255	-	File path of the car image
description	TEXT	-	-	Detailed product description

Table: tbl_orders

Field Name	Data Type	Size	Key	Description
order_id	INT	11	PK	Unique identifier for each order (Auto Increment)
user_id	INT	11	FK	References user_id in tbl_users
order_date	DATETIME	-	-	Date and time the order was placed
total_amount	DECIMAL	10,2	-	Total value of the order
order_status	VARCHAR	30	-	Pending / Confirmed / Delivered / Cancelled
payment_mode	VARCHAR	30	-	Mode of payment (Finance, Card, UPI, etc.)

Table: tbl_order_details

Field Name	Data Type	Size	Key	Description
order_detail_id	INT	11	PK	Unique identifier for each order line (Auto Increment)
order_id	INT	11	FK	References order_id in tbl_orders
car_id	INT	11	FK	References car_id in tbl_car
quantity	INT	11	-	Quantity of the car ordered
unit_price	DECIMAL	10,2	-	Price per unit at the time of order

Table: tbl_cart

Field Name	Data Type	Size	Key	Description
cart_id	INT	11	PK	Unique identifier for each cart entry (Auto Increment)
user_id	INT	11	FK	References user_id in tbl_users
car_id	INT	11	FK	References car_id in tbl_car
quantity	INT	11	-	Quantity added to the cart
added_date	DATETIME	-	-	Date and time item was added to cart

7. Conclusion

Car Showroom successfully demonstrates a complete e-commerce workflow for a car showroom business, integrating PHP and MySQL to deliver a functional, user-friendly, and scalable web application. The structured database design, clear data flow, and modular folder organization reflect sound software engineering practices suitable for academic evaluation and real-world extension.